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Review

Calibration of UV/Vis spectrophotometers: A review and comparison of different methods to estimate TSS and total and dissolved COD concentrations in sewers, WWTPs and rivers

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A B S T R A C T

UV/Vis spectrophotometers have been used for one decade to monitor water quality in various locations: sewers, rivers, wastewater treatment plants (WWTPs), tap water networks, *etc*. Resulting equivalent concentrations of interest can be estimated by three ways: *i)* by manufacturer global calibration; *ii)* by local calibration based on the provided global calibration and grab sampling; *iii)* by advanced calibration looking for relations between UV/Vis spectra and corresponding concentrations from grab sampling. However, no study has compared the applied methods so far. This collaborative work presents a comparison between five different methods. A Linear Regression (LR), Support Vector Machine (SVM), EVOlutionary algorithm method (EVO) and Partial Least Squares (PLS) have been applied on various data sets (sewers, rivers, WWTPs under dry, wet and all weather conditions) and for three water quality parameters: TSS, COD total and dissolved. Two criteria (r^2 and Root Mean Square Error *RMSE*) have been calculated - on calibration and verification data subsets - to evaluate accuracy and robustness of the applied methods. Values of criteria have then been statistically analysed for all and separated data sets. Non-consistent outcomes come through this study. According to the Kruskal-Wallis test and *RMSEs*, PLS and SVM seem to be the best methods. According to uncertainties in laboratory analysis and ranking of methods, LR and EVO appear more robust and sustainable for concentration estimations. Conclusions are mostly independent of water matrices, weather conditions or concentrations investigated.

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1. Introduction

For one decade, on-line and *in-situ* UV/Vis spectrophotometers have been used by researchers, engineers and practitioners (Langergraber et al., 2003; Gruber et al., 2006; van den Broeke et al., 2006; Hochedlinger et al., 2006) to measure certain pollutant concentrations on-line and with a high time-resolution.

Except in Qin et al. (2012) and Lourenço et al. (2008), only one type of spectrophotometer has been used: the spectro::lyser™ from s::can company; a submersible probe of cylindrical shape that measures optical absorbance from 200 to 750 nm with a step of 2.5 nm. Various versions and optical paths (distance between emitting and receiving cells, from 0.5 to 100 mm) are currently used and available depending to the expected range of concentrations in the investigated water matrix. The probe could be connected to a computer with a transmitter (con::nect) or to a devoted computer (con::stat, con::cube). The manufacturer of the probe is providing so called global calibrations to convert the measured spectra to so called equivalent concentrations for certain water quality parameters (TSS, COD, BOD, NO_X, *etc.*) depending on the water matrix. The equipment also includes a programmable self-cleaning valve system using pressurized air or water to ensure the necessary cleanness of the measurement window for reliable spectra recording.

According to the fact that water matrix vary spatially and temporally depending on various variables (Hochedlinger, 2005), a specific and local calibration is required for each new measurement site. Considering the wide panel of studies, calibration methodologies and criteria of performance, it is still hard to conclude about the most appropriate and relevant calibration approach. Main goals of this study are: *i)* to review the literature, *ii)* to select clear criteria to conclude about calibration quality, *iii)* to assess the performance of the most used calibration methods on several data sets, and *iv)* to analyse into detail the performance of these methods depending on the sites, the searched pollutants and weather conditions.

Due to the complexity of raw data (spectra) or equivalent concentrations (calculated with the global calibration provided by the manufacturer), there are a lot of methods available to evaluate pollutant concentrations resulting from a local calibration and criteria to assess their performances (presented in Section 2). Among existing methods, four have been selected: Linear Regression (LR) and Partial Least Squares (PLS) due to their occurrences and EVO and Support Vector Machines (SVM) due to promising results published in previous studies and briefly described in Section 3.2, applied to various data sets (see Section 3.1) and compared according to criteria (r^2, raw and normalized Root Mean Square Error) presented in the Section 3.2. Results are detailed and discussed for various groups of data sets (see Section 4). Conclusions, outcomes and perspectives of this study are finally highlighted.

2. Bibliographic review: existing methods and criteria

UV/Vis calibration methods are based on equivalent concentrations as linear or polynomial regressions, methods based on spectral regressions as PLS or machine learning methods as linear or generalized discriminant analysis, support or relevance vector machines, random forest and neural networks (Wolf et al., 2013). Local adjustments of the global calibration have been easily done (Torres, 2008; Caradot, 2012, Caradot et al., 2013). Linear regression is not necessarily the best fitting function by comparison with 2nd and 3rd order polynomials (Lepot et al., 2013). PLS methods are also used to find multi-linear functions between some wavelengths and concentration of water quality indicators (Rieger et al., 2004, 2006; Torres et Bertrand-Krajewski, 2008; Torres et al., 2013). Qin et al. (2012) published a comparative study between different PLS methods: *i)* a basic PLS, *ii)* the Uninformative Variable Elimination PLS (UVE-PLS) which eliminates the uninformative variables through a leave-one-out verification (Centner et al., 1996), *iii)* the Iterative Predictor Weighting PLS (IPW-PLS) which eliminates useless and redundant predictors in the PLS (Forina et al., 1999), *iv)* the Boosting PLS (Boosting-PLS), a sequentially additive regression which fits a learner in the residuals of the current function (Zhang et al., 2005) and *v)* the Boosting-IPW-PLS which combines the last two versions (Qin et al., 2012). Due to numerous unknown compounds involved in water matrix, the Lambert-Beer law is often not valid (Langergraber et al., 2003): linear methods are enough accurate to find correlation between spectra and certain concentrations of interest. Wolf et al. (2013) published a very exhaustive review of six machine learning methods to calibrate UV/Vis spectrophotometers: *i)* Linear Discriminant Analysis (LDA) fits a linear regression by maximizing the Fisher discriminant criterion, *ii)* Generalized Discriminant Analysis (GerDA) is the generalization of LDA to non-linear functions, *iii)* Random Forest (RF) solves complex classification and regression problems, *iv)* Multilayer Perceptrons (MLP) are feed-forward artificial neural networks, *v)* Support Vector Machines (SVM) are efficient for multi-classification problems and *vi)* Relevance Vector Machines (RVM) are Bayesian formulations of the classification problem with priors selected to encourage sparse representation. These machine learning methods can find non-linear regressions between spectra and concentrations.

Few of previously published studies present results about accuracy of calibration in case of sewer systems (Table 1). Two types of calibration have been found: Local if the calibration functions have been calculated with collected local grab samples, global in the opposite case by using grab samples from several locations. Mathematical methods (above mentioned) have been also

Table 1
A brief literature review of r^2 given by different calibration methods for TSS, COD total and dissolved in wastewater matrix (in italic, personal communications with authors and/or graphical readings, nc: non communicated).

Location	Type of calibration	Reference	TSS			Total COD			Dissolved COD		
			Number of samples	Number of wavelengths	r^2	Number of samples	Number of wavelengths	r^2	Number of samples	Number of wavelengths	r^2
WWTP	Local (PLS)	Langergraber et al. (2003)	28–24[a]	1[b]	nc - 0.83[a]	*21–25*[a]	1[b]	nc - 0.88[a]	*22–27*[a]	1[b]	nc - 0.72[a]
WWTP	Local (PLS)		28–24[a]	nc	nc - 0.95[a]	*21–25*[a]	nc	nc - 0.9[a]	*22–27*[a]	nc	nc - 0.91[a]
Paper mill WWTP	Local (PLS)	Langergraber et al. (2004a) [c]	–	–	–	*30–14*	1[b] (272–282 nm)[d]	0.27–0.59[c]	nc	1[b] (272–282 nm)[c]	0.27–0.59[c]
Paper mill WWTP	Global (PLS)		–	–	–	*30–14*	nc	0.58–0.76[c]	nc	nc	0.58–0.76[c]
Paper mill WWTP	Local (PLS)		–	–	–	*30–14*	nc	0.95–0.91[c]	nc	nc	0.95–0.91[c]
Experimental WWTP	Local (PLS)	Langergraber et al. (2004b)	*24*	nc	0.995	–	–	–	*14*	nc	0.9
WWTP	Local (PLS)	Rieger et al. (2004)	44	1[e] (378 nm)	0.326	–	–	–	12	1[e] (230 nm)	0.316
WWTP	Local (PLS)		44	7 (230–330 nm)	0.848	–	–	–	12	5 (250–340 nm)	0.905
WWTP	Local (LR)	Rieger et al. (2006) [f]	26	nc	0.83	–	–	–	–	–	–
WWTP	Global (PLS)		*24*	nc	0.83	35 - 26–47[d]	nc	0.23–0.83 - 0.93[d]	–	–	–
WWTP	Local (PLS)		26	nc	0.9	35 - 26–47[d]	nc	0.94–0.90 - 0.97[d]	–	–	–
WWTP (fuel company)	Local (PLS)	Lourenço et al. (2008) Salgado Brito et al. (2014)	*27–18*[a]	277 (224–500 nm)	0.84 - nc[a]	*26–19*[a]	36 (274–309 nm)	0.92–0.91[a]	–	–	–
WWTP	Global	Maribas et al. (2008)	–	–	–	nc	nc	0.84	–	–	–
WWTP	Spectral					nc	nc	0.94–0.97			
Sewer	Global (s::can)	Torres (2008)	16	nc	0.981	16	nc	0.911	–	–	–
Sewer	Local (LR)		16–8[a]	nc	0.975–0.998[a]	16–8[a]	nc	0.902–0.94	–	–	–
Sewer	Local (PLS)		16–8[a]	177 (290–730 nm)	0.996–0.996[a]	16–8[a]	210 (207.5–730 nm)	0.978–0.991	–	–	–
Sewer	Local (LR, EVO)	Gamerith et al. (2011b)	–	–	–	36	nc	0.91–0.94	–	–	–
Sewer	Local (SWLR[g])	Piro et al. (2012)	24	1 (730 nm)	0.87	24	1 (205 nm)	0.96	24	1 (200 nm)	0.88
Sewer & experimental WWTP	Local (PLS)	Qin et al. (2012)	163	1750	0.882	163	1750	0.815	–	–	–
Sewer & Experimental WWTP	Local (Boosting-PLS)		163	1750	0.884	163	1750	0.85	–	–	–
Sewer & experimental WWTP	Local (UVE-PLS)		163	279	0.922	163	205	0.837	–	–	–
Sewer & experimental WWTP	Local (IPW-PLS)		163	45	0.882	163	22	0.815	–	–	–
Sewer & experimental WWTP	Local (Boosting-IPW-PLS)		163	20	0.931	163	36	0.893	–	–	–
Sewer	Local (LR)	Caradot et al. (2013)	–	–	–	156	nc	0.86	–	–	–
River	Local (LR)		–	–	–	73	nc	0.3	–	–	–
River	Local (PLS)		–	–	–	73	nc	0.61	–	–	–
WWTP	Local (PLS)	Torres et al. (2013)	93	nc (550–735 nm)	(0.77–1)–(0.19–0.94)[a]	93	nc (200–735 nm)	(0.79–1)–(0.04–0.94)	93	nc (220–320 nm)	(0.88–1)–(0.88–1)
WWTP	Local (SVM)		93	nc (650–735 nm)	(0.69–0.86)–(0.50–0.88)[a]	93	nc (200–735 nm)	(0.79–0.94) – (0.59–0.96)	93	nc (220–320 nm)	(0.81–0.94)–(0.81–0.94)

[a] Calibration - Verification data subsets.
[b] Single wavelength linear regression result for comparison with PLS.
[c] Three paper mill industries.
[d] Influent - Effluent.
[e] Single wavelength linear regression result for comparison with PLS.
[f] Three different WWTP.
[g] Single wavelength linear regression.

reported. Most of methods presented by Wolf et al. (2013) haven't been used for TSS or COD concentration estimations in wastewater or river matrix. PLS is the most popular method. The following table summarizes data and results published in cited studies so far.

Except in Qin et al. (2012) and Torres et al. (2013), the used data set contain less than 50 samples. The PLS method seems to be the most popular and most of authors have used a local calibration. In most studies, information on selected wavelengths (number and positions) are often missing: published papers don't give access to the used part of spectra. Except in few studies, squared correlation coefficients r^2 of calibrated function are higher than 0.8. Only four papers have used a data set separation into a calibration and a verification data set (Langergraber et al. (2003), Laurenço et al. (2008), Torres (2008) and Salgado Brito et al. (2014)). Torres et al. (2013) have demonstrated that PLS method is more sensitive to outliers than SVM method. For COD total, main outcomes of the literature review are similar as the ones for TSS. The COD total seems to be well predicted by wavelengths in the UV range (Langergraber et al. (2004a), Laurenço et al. (2008) and Piro et al. (2012)). COD dissolved has been less studied so far and looks to be evaluated with a few wavelengths in the UV range (from 200 to 340 nm). Data sets used for this pollutant are small (often less than 30 samples).

The amount of available methods might be complicated for potential users who want to calibrate their own probes, especially when authors did not give used wavelengths, size of data sets and use different criteria to quantify accuracies of methods. Various criteria of performance have been used in the past. The coefficient of correlation has been the most used but looks to be slowly neglected in recent articles. Errors of prediction (i.e. differences between laboratory and calculated values) have been often used to evaluate accuracy of regression patterns (Table 2). Salgado Brito et al. (2014) presented a good overview of published criteria with their advantages and disadvantages. Notations used in equations presented in Table 2 have been harmonized: M is the number of samples, p_i is the laboratory concentration (in mg/L), P_i is the predicted concentration by the regression function (in mg/L), $\overline{p}$ and respectively $\overline{P}$ are the average values of laboratory and respectively predicted concentrations (in mg/L).

Table 2 demonstrates two unfortunate findings. An equation may have different names: the fourth equation could be named as RMSE or RMSEP. Otherwise, a same name could be linked to two different equations: RMSE is linked to the fourth and the fifth equations. Most authors, as mentioned in previous studies, used only few of them to characterize accuracies of methods: it is not always easy to compare published methods if calculated criteria are not the ones used by other authors, if the number of sample of the data set and of selected wavelength(s) are also not given. Few remarks and comments on criteria are given at the end of this paper.

3. Materials and methods

3.1. Materials: construction of data sets

The data sets used in this study are considered as materials. Sites, probes, laboratory analyses, calculation of uncertainties are briefly described hereafter for each data set. The UV/Vis probes used were spectro::lysers from the s::can company. Based on the measured absorbance at different wavelengths, so called equivalent concentrations are calculated by means of the manufacturer's global calibration (a relation between selected wavelengths and concentrations of pollutants) for COD, TSS, nitrate, etc. A detailed description of the sensor can be found e.g. in Langergraber et al. (2003). Three different measurement conditions were considered in this study: i) dry weather, named DW, ii) wet weather (influence of stormwater runoff) named WW and iii) all weather (combination of dry weather and wet weather data sets) named AW.

Table 2
Review of criteria used to quantify accuracy of regression functions.

Nr.	Equation	Name(s)	Reference(s)
C1	$\left\{\frac{[\sum_{i=1}^{M}(P_i-\overline{P})\times(p_i-\overline{p})]}{\sqrt{\sum_{i=1}^{M}(P_i-\overline{P})^2\times\sum_{i=1}^{M}(p_i-\overline{p})^2}}\right\}^2$	Coefficient of correlation (r^2)	Qin et al. (2012) See Tables 1–3
C2	$max(P_i - p_i)$	Maximum prediction error (MaxE)	Qin et al. (2012)
C3	$min(P_i - p_i)$	Minimum prediction error (MinE)	Qin et al. (2012)
C4	$\sqrt{\frac{\sum_{i=1}^{M}(P_i-p_i)^2}{M}}$	Root Mean Square Errors of Prediction (RMSEP)	Lourenço et al. (2008) Qin et al. (2012)
		Root Mean Square Errors (RMSE)	Hochedlinger et al. (2006) Lepot et al. (2013) Torres et al. (2013) Salgado Brito et al. (2014) Caradot et al. (2013)
C5	$\sqrt{\frac{\sum_{i=1}^{M}(P_i-p_i)^2}{M-1}}$	Root Mean Square Errors of Cross-Verification (RMSECV)	Lourenço et al. (2008)
C6	$\sqrt{\frac{\sum_{i=1}^{M}(P_i-p_i)^2}{M-2}}$	Residual standard deviation (s_X)	Rieger et al. (2006)
C7	$100\times\frac{\sqrt{\frac{\sum_{i=1}^{M}(P_i-p_i)^2}{M}}}{\overline{p}}$	Relative Root Mean Square Errors ($RMSE_{REL}$)	Salgado Brito et al. (2014)
C8	$100\times\frac{\sqrt{\frac{\sum_{i=1}^{M}(P_i-p_i)^2}{M-2}}}{\overline{p}}$	Coefficient of Variation (CV)	Rieger et al. (2006)
C9	$\frac{\sqrt{\sum_{i=1}^{M}(P_i-p_i)^2}}{\sqrt{\sum_{i=1}^{M}(p_i-\overline{p})^2}}$	Root mean squared error standard deviation ratio of observation (RSR)	Moriasi et al. (2007) Salgado Brito et al. (2014)
C10	$1-\frac{\sum_{i=1}^{M}(P_i-p_i)^2}{\sum_{i=1}^{M}(p_i-\overline{p})^2}$	Nash-Sutcliffe coefficient of efficiency (NSE)	Nash and Sutcliffe (1970) ASCE (1993) Moriasi et al. (2007) Salgado Brito et al. (2014)
C11	$\frac{\sum_{i=1}^{M}(p_i-P_i)}{\sum_{i=1}^{M}p_i}$	(PB_{ias})	ASCE (1993) Gupta et al. (1999) Salgado Brito et al. (2014)
C12	$\frac{\max(p_i)-\min(p_i)}{\sqrt{\frac{\sum_{i=1}^{M}(P_i-p_i)^2}{M}}}$	Range to Error Ratio (RER)	Sarraguça et al. (2011) Salgado Brito et al. (2014)

3.1.1. Austrian data sets: Graz (sewer) and Linz (WWTP)

All grab samples collected in Graz were taken at the combined sewer overflow (CSO) R05 measurement station located at the outlet of the urban catchment Graz West CSO R05 (20 000 inhabitants) drained by a combined system (Gruber et al., 2005; Hochedlinger, 2005; Gamerith, 2011a). The *in-situ* spectrophotometer (optical path length: 5 mm) has been installed directly in the sewer, attached under a floating pontoon. Three data sets (dry, wet and all weather conditions) have been collected by two types of sampling campaigns: automatic grab sampling (with an automatic peristaltic sampler) in 1 h interval during dry weather and manual grab sampling during rain events. In total 85 dry and 83 wet weather samples had been collected in the period from July 2003 until September 2009. Each sample corresponds to a composite sample over the filling time period of one sampler bottle which takes between 3 and 5 min. At least three spectra have been recorded during this filling period, analysed manually to detect erroneous recordings and then averaged for calculating the derived water quality parameters (Gamerith et al., 2011c). The collected composite samples have been homogenized by means of an Ultraturrax mixer before laboratory analyses were done. The three following Graz data sets have been used in this study: TUG-Graz-DW (dry weather), TUG-Graz-WW (wet weather) and TUG-Graz-AW (all weather conditions).

Data sets from Linz were taken at inlet and outlet of the primary clarifiers installed at the Linz-Asten WWTP (Hofer et al., 2013). At both sites the optical path length of the two spectrophotometers was 5 mm. Grab samples and corresponding spectra were collected during one 24-h sampling campaign for both sites. The collected grab samples have been refrigerated and homogenized by means of an Ultraturrax mixer before laboratory analyses were done. The three following Linz data sets have been used in this study: TUG-Linz-pcIN (24 samples at the inlet of the primary clarifier), TUG-Linz-pcOUT (24 samples at the outlet of the primary clarifier) and TUG-Linz-pcALL (48 samples from both locations). No wet weather sampling was done at this site.

Laboratory analyses have been performed according to standard methods ISO 15705 (2002) for total and filtered COD and DIN 38409-2 (1987) for TSS. Standard uncertainties have been assumed equal to 10% of the laboratory values (Lombard et al., 2010).

3.1.2. Colombian data sets: El Salitre (sewer), river (river) and CCS (sewer)

Three data sets have been built with Colombian samples and one spectrophotometer (optical path length: 5 mm). 27 samples have been collected during dry weather in the Canal de Pontificia Universidad Javeriana to build the data set called hereafter PUJ-CCS (dry weather). This open channel collects rain and domestic waters and presents a high concentration of vegetal organic matter (due to the vegetation along both sides of the channel). The data set PUJ-WW (wet weather) contains 41 samples collected during wet weather in the Gibraltar pumping station located in the South-East part of Bogota along the El Salitre river. A third series of 21 samples has been manually collected in the Rio Arzobispo urban river (length of approximately 1.8 km) during rain events: these samples constitute the PUJ-RW-WW (wet weather) data set.

Laboratory analyses have been performed by triplicates according to the methods 2540 for TSS and 5220 for total and filtered COD (APHA, 2012). Standard uncertainties (including analysis and sampling uncertainties) have been estimated according to the method presented in Torres-Abello (2011) based on international standards (ISO GUIDE 98-1, 2009).

3.1.3. French data sets: Fontaines-sur-Saone (WWTP)

The spectrophotometer used in Fontaines-sur-Saône had an optical path length of 2 mm. Samples were collected at the Fontaines-sur-Saône WWTP, immediately downstream the pre-treatment steps (screen bars, sand, grit and grease removal) and were transported directly to the laboratory without any processing or conditioning. The WWTP collects wastewater from a combined sewer system serving about 30,000 inhabitants.

Three data sets (dry, wet and all weather conditions) were collected during 7 non-consecutive days in 2011. For each 1 L grab sample, between 10 and 15 spectra have been recorded by installing the spectrophotometer in a closed pipe with the sample circulated by means of a peristaltic pump. Pollutant concentrations were measured by triplicate standard laboratory analyses for TSS (NF T90-105-2, 1997) and total and filtered COD (ISO 15705, 2002). Average concentrations, standard uncertainties, pre-treatment of spectra and outliers detection were carried out according to the methods presented in Lepot et al. (2013). Three data sets have been built with theses samples: INSA-DW (dry weather), INSA-WW (wet weather) and INSA-AW (all weather conditions).

3.1.4. German data sets: Container (sewers) and river (river)

The Container CSO monitoring station was installed on a major overflow sewer in the city center of Berlin (Caradot et al., 2013). Water quality was measured in a bypass fed by a peristaltic pump, using a UV/Vis spectrophotometer with an optical path length of 5 mm. 59 samples have been taken during 20 CSO events to build the KWB-WW data set.

The river monitoring station (Caradot, 2012) was equipped with a spectrophotometer with an optical path length of 35 mm attached on metal supports and immersed directly into the river 500 m downstream the CSO outlet. Samples were taken during CSO events by automatic peristaltic samplers if the flow exceeded a given threshold (Container CSO outlet) or if the COD_{eq} concentration exceeded a given threshold (River). Samples have also been taken on a regular basis during dry weather. 51 samples have been collected during 15 CSO events to build the KWB-RW data set.

COD has been analysed on site using Hach Lange test cuvettes (ISO 15705, 2002) and TSS have been analysed in laboratory (EN 12880, 2000). For both sites, COD and TSS uncertainties have been estimated to 10% by propagating uncertainties (Lombard et al., 2010) of sampling (5%) and laboratory analyses (5%).

3.1.5. Summary of main characteristics of each data set

The main characteristics of each data set are given in Appendix A (tables A1, A2 and A3) and the data sets are given in the Supplementary Materials.

The data sets include various water matrix (sewer, WWTP, river) and weather conditions. Their sizes are larger than most of the data sets found in the literature review (Table 1). Concentration ranges overlap with usually observed values for similar water matrix (e.g. in Ellis et al., 2005).

As an example, Table 3 is an extract of Appendix A with Fontaines-sur-Saône INSA_AW data sets: it shows the number of samples, weather conditions, location and some basic statistics. As shown in Table 3, tables in Appendix A allow the comparison of complete, calibration and verification data subsets by means of the values given in the last three columns: minimum, mean, maximum, median values and standard deviation of pollutant concentrations. Each data set has been divided in two subsets: the first 70% of the samples were used for calibration and the last 30% for verification. The last three columns (all, calibration and verification samples) highlight that: i) concentration ranges are consistent with common ranges and ii) calibration and verification subsets are not significantly different.

Table 3
Main characteristic of INSA_AW data set for TSS (2nd row), total (3rd row) and filtered (4th row) COD. SD: Standard Deviation.

Name (number of samples)	Weather	Location	Min-mean-max Median − SD (mg/L) all samples	Min-mean-max Median − SD (mg/L) calibration samples	Min-mean-max Median − SD (mg/L) verification samples
INSA-AW (135)	All	WWTP	26–157–322 160–78	26–162–322 166–78	31–143–311 146–76
INSA-AW (136)	All	WWTP	59–333–717.5 290–180	74–379–717.5 387–184	59–227–570 207.5–114
INSA-AW (138)	All	WWTP	19–109–283 76–74	26–130–283 141–78	19–60–160 55–27

3.2. Methods

3.2.1. Sensor calibration methods

Two approaches have been investigated to calibrate UV/Vis spectrophotometers: i) methods based on equivalent concentrations resulting from the global calibration provided by the manufacturer (LR method) and ii) methods based on spectral data (EVO, PLS and SVM methods).

The detailed description (equations, algorithms, etc.) of all above methods is not part of this paper: the reader should refer to the references given in the following paragraphs which briefly introduce each method.

3.2.1.1. LR method. The linear regression (LR) is a very basic method described in ISO 8466-1 (1990). Laboratory measurements are correlated with in-situ estimations of equivalent concentrations resulting from the global calibration provided by the manufacturer. The linear function obtained by least squares minimization is then applied to correct online estimations of equivalent concentrations according to equation (1):

$$[P] = b_0 + b_1 \times C_{eq} \quad (1)$$

where *[P]* is the reference concentration (mg/L) measured in the laboratory according to standard methods, b_0 and b_1 are coefficients and C_{eq} is the equivalent concentration (mg/L) resulting from the manufacturer global calibration.

3.2.1.2. Spectral methods. The generic equation of spectral functions is given by equation (2):

$$[P] = b_0 + \sum_{i=1}^{n} b_i \times A_i \quad (2)$$

where b_0 and b_i are coefficients, A_i is the absorbance (m^{-1}) at the selected *i*th wavelength and *n* is total number of selected wavelengths.

3.2.1.3. EVO method. The EVO method, presented in detail in Muschalla (2008), is based on a global optimiser and on evolutionary strategies which allow multi-objective optimisation. This method has been developed by Muschalla (2006) and successfully applied in calibration studies (e.g. Gamerith, 2011a).

3.2.1.4. SVM methods. Support Vector Machine regression and classification are very useful in order to detect patterns in complex and non-linear data (Schölkopf and Smola, 2002). The main problem solved by SVM is the adjustment of a function describing a relationship between objects *X* and the answer *Y* using the data set *S* (López-Kleine and Torres, 2014). For calibration, the SVM method was applied in combination with a leave-one-out cross-verification procedure (LOOCV) using as objective function the squared differences between observed and SVM regression estimated data in order to determine the most predictable SVM method (Torres et al., 2013).

3.2.1.5. PLS method. The Partial Least Squares (PLS) method (Abdi, 2003; Bertrand, 2005; Dos Santos Climaco Pinto, 2009) predicts a variable *Y* (pollutant concentrations) from a matrix *X* of partially dependent variables (absorbances from wavelengths spectra) based on their common structure. The PLS method used in this study has been already published in Torres and Bertrand-Krajewski (2008) and includes a leave-one-out cross-verification procedure (LOOCV).

3.2.2. Comparison and assessment of applied calibration methods

3.2.2.1. Applied criteria: r^2, RMSE and normalized values. Criteria are necessary to evaluate and compare the performance of sensor calibration methods, for both calibration and verification. In case of calibration, the criteria indicate the ability of the method to represent concentrations measured in calibration samples. This refers to method accuracy. In case of verification, the criteria indicate the ability of the method to estimate concentrations in samples which were not used to establish the calibration function. This refers to method robustness, i.e. its ability to provide estimations with similar level of accuracy as for calibration.

The choice of adequate performance criteria is not obvious, as revealed by Table 2 which contains 12 equations for various criteria. In this work, we use five criteria for both calibration and verification. Two of them are listed in Table 2: we consider them as relevant for our work and they allow comparison with previously published papers. The three other ones are normalised versions of the two first ones: they facilitate the comparison and ranking of methods for data subsets corresponding to various water matrix and concentration ranges.

The five criteria are:

- square of the correlation coefficient r^2 (-): criterion C1 in Table 2.
- root mean square error *RMSE* (mg/L): criterion C4 in Table 2.
- normalized correlation coefficient:

$$Nr^2_{01} = \frac{r^2 - r^2_{min}}{r^2_{max} - r^2_{min}} \quad (C13)$$

where r^2_{min} and r^2_{max} are respectively the minimum and maximum values of r^2 obtained with the four methods for a given data subset. Nr^2_{01} varies from 0 to 1 for the worst to the best model according to the r^2 criterion.

- normalized *RMSE* (-):

$$NRMSE_{01} = \frac{RMSE_{max} - RMSE}{RMSE_{max} - RMSE_{min}} \tag{C14}$$

where $RMSE_{min}$ and $RMSE_{max}$ are respectively the minimum and maximum values of *RMSE* obtained with the four methods for a given data subset. $NRMSE_{01}$ varies from 0 to 1 for the worst to the best model according to the *RMSE* criterion.

- Concentration range normalized *RMSE* (-):

$$NRMSE = \frac{RMSE}{p_{max} - p_{min}} \tag{C15}$$

where p_{min} and p_{max} are respectively the minimum and maximum pollutant concentrations in a given data subset.

3.2.2.2. Performance and comparison tests. The distributions of the above criteria have been compared to theoretical distributions (normal, log-normal, *etc.*) according to groups of data sets presented in Table 4. Preliminary investigations (not shown here) were carried out to choose parametric or non-parametric tests to compare the calibration methods. The objective is to evaluate if some calibration methods appear as more appropriate for some groups of data.

Three tests have been applied to the 10 groups of data:

- The Kruskal-Wallis test: the *p*-value of the null hypothesis has been calculated for every group of data sets, for r^2 and *RMSE* criteria and for calibration and verification data subsets. If the *p*-value is lower than 0.01, distributions are significantly different according to the examined criteria and if the *p*-value is higher than 0.01, methods perform equivalently.
- A comparison between positions: according to r^2 and *RMSE* criteria (C1 and C4), the four methods applied in this study have been ranked from one (best case) to four (worst case). As normalized criteria (C13 to C15) do not change the relative ranking position, comparison is done only for C1 and C4.
- Another test has been applied to account for uncertainties in laboratory measurements. For each data set, one considers that calibration methods give equivalent results if the differences between *RMSE* values are below a given threshold: 10 mg/L for TSS and 20 mg/L for COD, according to usual uncertainties in laboratory analyses. On the other hand, if the differences are higher than the thresholds, one can conclude that the methods give significantly different results according to the accuracy of laboratory analyses.

In order to synthetize the results of these tests, the consistency ratio has been calculated for each method and for each group of data sets. It is the ratio between the number of data sets for which the *RMSE*s difference between two methods is lower than the threshold and the total number of data sets in the same group. The lowest the consistency ratio between two methods, the highest the difference between the *RMSE* values of the two methods. These ratios have been calculated for both calibration and verification data subsets.

Table 4
The ten groups of data sets used in this study.

Group nr.	Data sets sorted by	Category/groups (number of data sets)
G1	No sorting	All data (42)
G2	Pollutants	TSS (14)
G3		COD total (14)
G4		COD filtered (14)
G5	Sites	Sewers (18)
G6		WWTP (18)
G7		River (6)
G8	Weather conditions	Dry weather (18)
G9		Wet weather (18)
G10		All weather (6)

4. Results and discussion

All results for each data set, pollutant, method and criterion are given in Appendix B which includes Table B1 for TSS, Table B2 for total COD and Table B3 for filtered COD.

Due to space limitation in this paper, results are presented with some detail only for groups of data sets G1 and G2. A global analysis is then given for all groups defined in Table 4.

4.1. Group G1: all data sets

Fig. 1 shows the ranking of the tested methods based on position occurrences (bars) and average positions (vertical lines) for both *RMSE* (upper plots) and r^2 criteria (lower plots), respectively for calibration data sets (left column) and verification data sets (right column). The position occurrence indicates how often a given calibration method obtained each position, from the best (position 1) to the worst (position 4).

Let's consider first the calibration results in Fig. 1. For *RMSE* (top left graph), PLS and SVM are respectively ranked first and second best methods, i.e. they give the lowest *RMSE* values 23 times and 15 times respectively, and the second lowest RMSE values 24 and 16 times respectively (among 42 available data sets). On the contrary, LR and EVO are most frequently ranked as the third and fourth best methods. For example, EVO is ranked fourth method for 38 data sets among 42. EVO is even never ranked as the best method for any of the 42 data sets. For r^2 (down left graph), the occurrences are rather similar compared to *RMSE* for PLS and SVM (22 and 17 times ranked first respectively), and different for LR and EVO which are more equivalently ranked third and fourth best methods. Considering the mean position, for both *RMSE* and r^2, the vertical lines for PLS and SVM are close each other and significantly different from the vertical lines for LR and EVO. Clearly, PLS and SVM perform better for calibration than LR and EVO, when considering both *RMSE* and r^2 for all data sets together.

Looking at the verification results (right graphs in Fig. 1), the methods appear less distinguishable than for calibration. For *RMSE* (top right graph), SVM is first ranked 14 times, LR 13 times, EVO 8 times and PLS 7 times. PLS is ranked as the fourth best method the most frequently (22 times). Considering mean values, the hierarchy for verification is SVM, LR, EVO and, a little bit behind, PLS. For r^2 (down right graph), verification results are globally similar to those obtained for *RMSE* and again significantly different from calibration results. The hierarchy, in decreasing order, is LR, EVO, SVM and PLS.

These results indicate that the most frequently best method for calibration is not necessarily the most frequently best method for verification, *i.e.* for predicting new equivalent concentrations. The most sophisticated methods (SVM and PLS) show good performance to represent calibration data sets. Their algorithms are powerful enough to represent accurately the observed data with their variability or dispersion. But it seems that they are too overfitted (especially PLS) to the calibration data sets: they lose performance for verification and are thus less robust than a more simple method like LR. Penalty criteria can be applied on the number of selected wavelength to improve the robustness of PLS model while keeping a good accuracy (Lepot, 2012).

Fig. 2 presents the consistency ratios for all data sets (group G1). Let's consider the EVO method for calibration (left graph). Its own

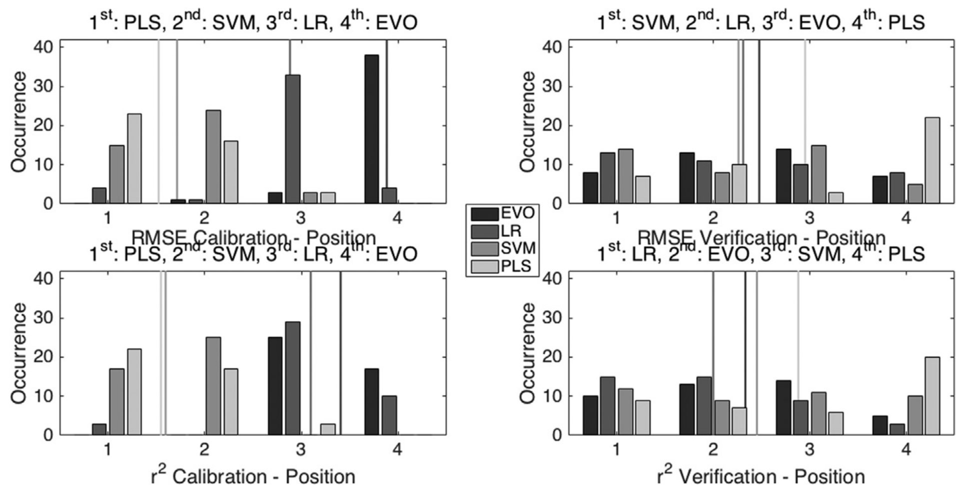

Fig. 1. Summary of ranking analysis for all data sets (ranking: 1 = best, 4 = worst).

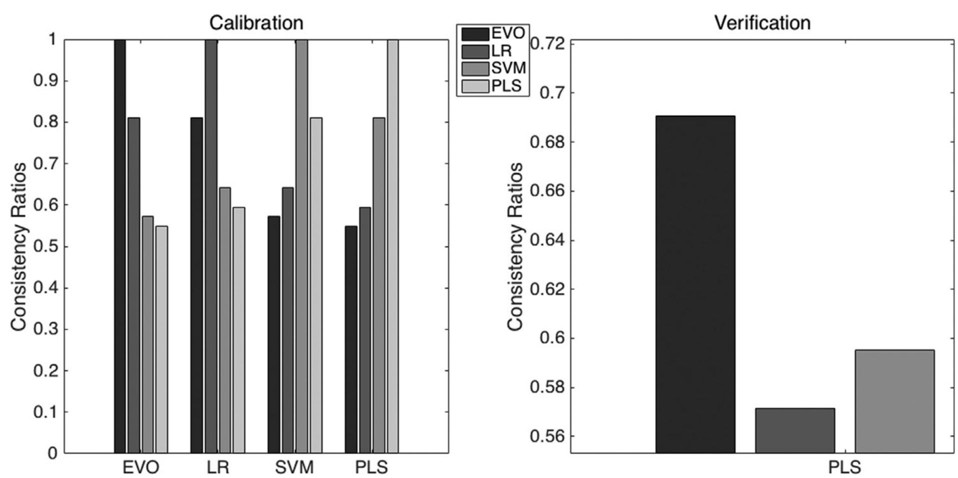

Fig. 2. Consistency ratios for all 42 data sets (left: calibration results, right: verification results).

consistency ratio is equal to 1, by definition. LR has a consistency ratio of 0.81 compared to EVO, SVM and PLS have consistency ratios of 0.57 and 0.55 respectively. This indicates that, globally compared to EVO, LR provides calibration results which, according to the defined thresholds, are not significantly different, whereas SVM and PLS are much different. If one considers LR as the reference, EVO has a consistency ratio of 0.81, and SVM and PLS have ratios respectively equal to 0.64 and 0.59. If PLS is the reference, EVO has a consistency ratio of 0.54, LR and SVM have ratios respectively equal to 0.59 and 0.81. For the group G1 calibration, EVO and LR provide rather similar results, while SVM and PLS are both similar to each other and significantly different from EVO and LR.

For verification data subsets (right graph in Fig. 2), when considering PLS as the reference, EVO, LR and SVM have consistency ratios of 0.69, 0.57 and 0.59 respectively. The other blocks of bars (i.e. choosing EVO, LR and SVL as references) indicates that consistency ratios for verification globally range between 0.55 and 0.75. This confirms that differences in *RMSE* results for group G1 are globally more significant in verification than in calibration.

In addition, *p*-values given by the Kruskal-Wallis test are low for calibration (5.10^{-18} and 0.0012 respectively for r^2 and *RMSE*) and below 0.01, indicating that results are significantly different for the various methods. On the contrary, *p*-values are high above 0.01 for verification (0.578 and 0.6876 respectively for r^2 and *RMSE*), which means that the different methods cannot really be ranked with significant differences.

Table 5
Average standard uncertainties for laboratory analyses. Calculated values are given first, and 10% estimates are given between brackets; nc indicates that uncertainties were not calculated.

Data set	Average standard uncertainties in laboratory analyses (mg/L)		
	TSS	Total COD	Filtered COD
INSA-AW	7.4 (15.6)	6.4 (33.3)	5.0 (10.9)
INSA-DW	9.5 (16.5)	7.4 (43.5)	6.1(15.3)
INSA-WW	7.0 (16.6)	6.5 (25.1)	5.2 (5.8)
PUJ-CCS	3.9 (8.1)	11.7 (25.1)	10.6 (27.1)
PUJ-WW	7.9 (16.3)	9.8 (50.9)	9.2 (35.6)
PUJ-RW-WW	2.5 (2.8)	8.8 (5.2)	9.6 (5.6)
KWB-RW	nc (2.8)	nc (3.8)	nc (2.2)
KWB-WW	nc (16.3)	nc (26.3)	nc (5.9)
TUG-Graz-AW	nc (20.4)	nc (51.9)	nc (19.3)
TUG-Graz-DW	nc (21.1)	nc (69.7)	nc (21.2)
TUG-Graz-WW	nc (19.7)	nc (33.6)	nc (6.5)
TUG-Linz-pcIN	nc (31.7)	nc (69.2)	nc (18.2)
TUG-Linz-pcOUT	nc (9.8)	nc (43.2)	nc (18.2)
TUG-Linz-pcALL	nc (20.7)	nc (56.2)	nc (18.2)

The consistency ratios depend on the defined thresholds: 10 mg/L for TSS and 20 mg/L for COD. These thresholds were defined as default values to account for uncertainties in laboratory analyses. Indeed, uncertainties in laboratory analyses were only assessed specifically for the INSA and PUJ data sets (accounting for uncertainties in analytical method and sub-sampling for triplicated analyses, Bertrand-Krajewski et al., 2007). No specific uncertainty assessment was done for the other data sets, which consider a relative standard uncertainty of 10% for all TSS and COD concentrations. Table 5 displays uncertainties in concentrations both as calculated (for INSA and PUJ data sets) and as 10% estimates for all data sets. 10% estimates always overestimate calculated uncertainties. However, to allow the comparison of all data sets in this paper in a simplified way, the 10% estimates, converted into 10 mg/L for TSS and 20 mg/L for COD, were used in the calculation of consistency ratios. The consequence is that consistency ratios are slightly overestimated: the differences between methods would probably be more significant if local specific uncertainties would have been applied instead of 10% estimates.

4.2. Groups G2: TSS data sets

In total 14 different TSS data sets are available. Fig. 3 gives the results of the ranking analysis. For calibration, PLS and SVM perform better, with similar results for r^2 and *RMSE* (only one vertical line is visible on the graphs). For *RMSE*, LR appears better than EVO, while they are almost equivalent for r^2. For verification, EVO, LR and SVM are close each other and slightly better than PLS. As for the group G1, SVM and PLS are the best methods for calibration, but EVO and LR are better for verification.

Kruskal-Wallis *p*-values for calibration are 0.03 for r^2 and 0.11 for *RMSE*, and for verification 0.82 for r^2 and 0.93 for *RMSE*. This means that differences between the methods are not very significant when considered globally. This test leads to less obvious conclusions than the simple ranking comparison.

Fig. 4 shows two radar plots for *RMSE* results. The top plot displays the results for calibration. The PUJ-WW, TUG-Graz-WW, TUG-Graz-DW and TUG-Graz-AW data sets show higher *RMSE* values ranging between 65 and 111 mg/L (except 7.23 mg/L for TUG-Graz-WW with PLS), which indicates that calibration functions are of lower accuracy than for the ten other data sets with RMSE values ranging from almost 0 to 60 mg/L (median value = 31 mg/L). For verification (down plot), the three Graz data sets again show the highest *RMSE* values between 72 and 226 mg/L. The other data sets have much lower *RMSE* values in the range 6–106 mg/L, with a median value of 38 mg/L. It confirms that verification performance are lower than calibration performance.

Fig. 5 presents the consistency ratios for TSS data sets. For calibration, if PLS is taken as reference, the consistency ratios are 0.21, 0.14 and 0.50 for EVO, LR and SVM respectively. Such low values indicate that the results are significantly different for the various methods. For verification, again with PLS as reference, consistency ratios are 0.36, 0.21 and 0.36 for EVO, LR and SVM respectively. Again, the results vary between the methods.

4.3. Global analysis of the ten groups of data sets

Groups G2 to G4, G5 to G7 and G8 to G10 (see Table 4) were analysed separately to identify possible differences between pollutants, sites (i.e. water matrix) and weather conditions for the tested calibration methods. Detailed results for all five criteria are given in Appendix B but are not presented in this paper due to space limitations. They reveal some differences and some variability, but these differences are not systematic and strong enough to conclude that some calibration methods are better for some cases (pollutants, sites or weather conditions) than for other ones.

For the final global comparison, a synthesis of the ranking results is given in Table 6. This synthesis has been established manually and accounts for all five performance criteria presented in Section 3.2.2 and given in Appendix B. For both calibration and verification, the four methods are given on each line from the best one (left side) to the worst one (right side), with comparative symbols (~for similarity or very small superiority, > for superiority and ≫ for great superiority). For example, calibration for the group G3 indicates PLS ~ SVM ≫ EVO ~ LR. This means that PLS performs rather similarly to or slightly better than SVM, that SVM performs much better than EVO which itself is a slightly better than LR.

Table 6 shows that PLS and SVM are the two most frequently best performing methods for all groups in case of calibration and that they perform better or much better than EVO and LR, LR showing the worst performance. On the contrary, for verification, EVO and LR are globally the most frequently best performing methods, with a very small superiority compared to SVM, which itself performs a little better than PLS.

The ranking analysis for all groups considered separately show some variability but the results presented above with all data sets

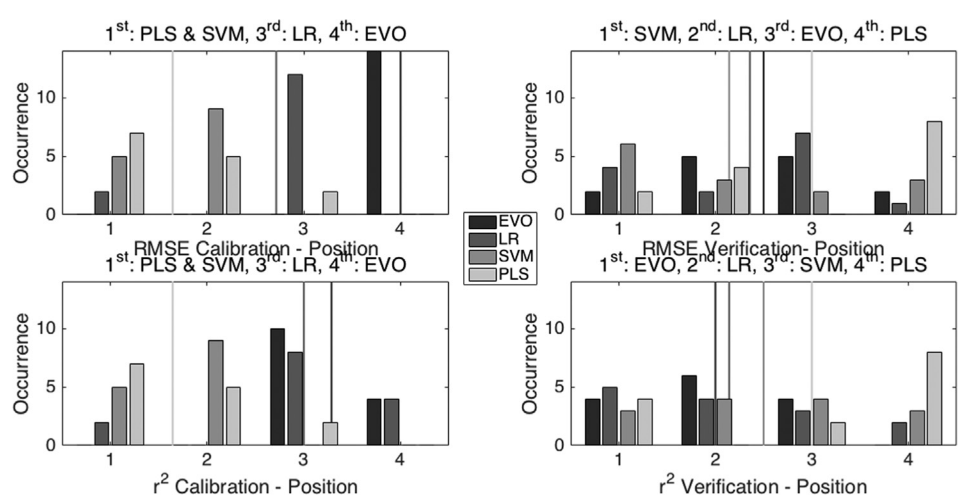

Fig. 3. Summary of ranking analysis for the 14 TSS data sets (ranking: 1 = best, 4 = worst).

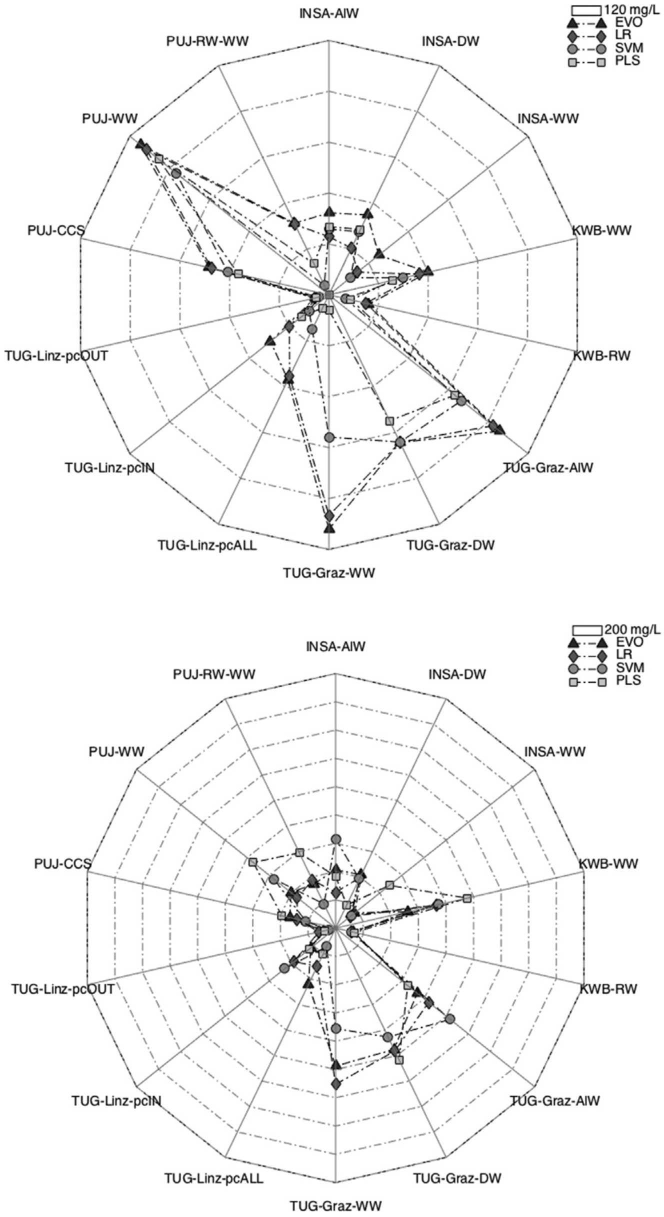

Fig. 4. *RMSE* radar plots for TSS data sets (top: calibration; down: verification).

together (group G1, Section 4.1) are confirmed: the most sophisticated methods (PLS and SVM) perform very well for calibration, as they are able to better capture the complexity and the diversity of data included in the calibration data sets. However, when the resulting calibration functions are used to predict new values of concentrations from spectra not used for calibration (verification data sets), the performance of PLS and SVM decreases; in most cases EVO and to a lesser extent LR perform better. LR and EVO, with a lower degree of complexity compared to PLS and SVM, are more robust for prediction of concentrations.

5. Conclusions and perspectives

The two main objectives of this paper were:

- to review the methods used to calibrate UV/Vis spectrophotometers for various water matrix;
- to compare four calibration methods applied with various data sets covering different types of water, sites, and weather conditions.

Regarding the first objective, the main conclusions are:

- Comparisons of global and local calibration methods by various authors demonstrated that local calibration is always the most accurate method (according to the r^2 criteria) to account for water matrix specific characteristics and is thus required.
- Most published applications used the PLS method to establish a relationship between spectra and concentrations.
- PLS adaptations and SVM methods are the most recent applied methods to estimate concentrations and seem promising as they perform better than simple methods for calibration.
- Data sets, concentration ranges, and statistics of data sets and applied methods are rarely given in the published studies.
- There is no general consensus about performance criteria to be used to evaluate the calibration methods, which makes comparison difficult.

Regarding the comparison of four calibration methods (LR, EVO, SVM and PLS) applied to 14 large data sets representing different water matrix and sites, pollutants and weather conditions in four countries, the main information and results are the following ones:

- The comparison of the methods was based on r^2 and *RMSE* criteria, completed with three normalised criteria (Nr^2_{01}, *NRMSE*

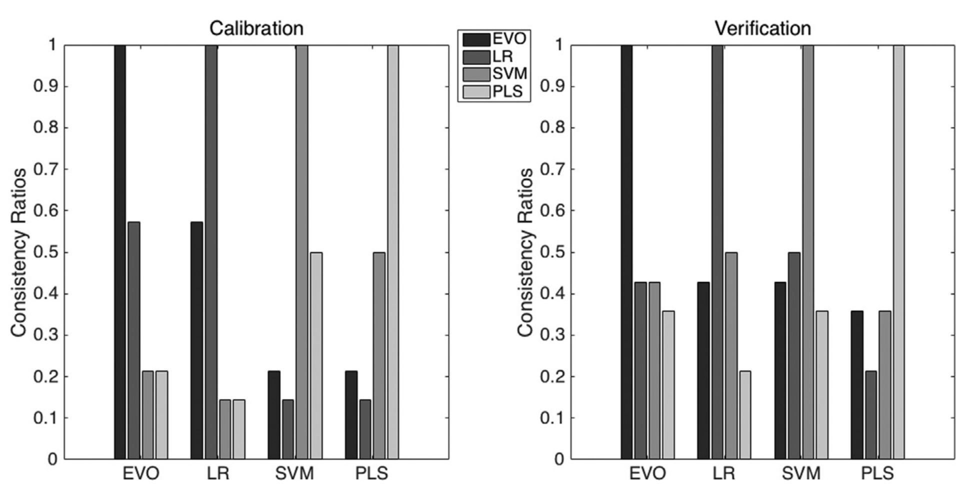

Fig. 5. Consistency ratios for all TSS data sets (left: calibration results, right: verification results).

Table 6
Summary of analysis performed on the 10 groups of data (see Table 4).

Group nr.	Ranking for calibration	Ranking for verification
G1	PLS ~ SVM > EVO ~ LR	EVO ~ LR ~ SVM > PLS
G2	PLS > SVM ≫ EVO ~ LR	EVO ~ LR > PLS ~ SVM
G3	PLS ~ SVM ≫ EVO ~ LR	EVO ~ LR ~ SVM > PLS
G4	PLS ~ SVM > EVO ~ LR	LR > EVO ~ SVM > PLS
G5	PLS > SVM ≫ EVO ~ LR	EVO ~ LR ~ SVM > PLS
G6	PLS ~ SVM ≫ EVO ~ LR	EVO ~ LR ~ SVM ~ PLS
G7	SVM > PLS ≫ EVO ~ LR	EVO ~ LR ~ SVM ~ PLS
G8	PLS ~ SVM > EVO ~ LR	SVM > EVO ~ LR > PLS
G9	PLS ~ SVM > EVO ~ LR	EVO ~ LR ~ SVM > PLS
G10	PLS ~ SVM > EVO ~ LR	EVO ~ LR ~ SVM ~ PLS

and $NRMSE_{01}$) to account for different ranges of concentrations in the various data sets.
- PLS completed with cross-verification outperforms the other methods for calibration.
- SVM appears as the second best method for calibration.
- The most sophisticated methods (PLS and SVM) perform very well for calibration and much better than EVO and LR. However, when the resulting calibration functions are used to predict new values of concentrations from spectra not used for calibration (verification data sets), the performance of PLS and SVM decreases. For prediction, EVO and to a lesser extent LR perform better: they are more robust for prediction of concentrations than PLS and SVM.

The choice of a calibration method is a trade off between accuracy, robustness and mathematical complexity. From an operational point of view for practitioners in charge of monitoring urban water systems (sewers, WWTP, rivers), the authors suggest that a simple method like LR is likely the most appropriate option as predictions of concentrations are the most robust ones if a sufficiently large and representative data set (regarding the coverage of concentration range and weather conditions) is locally collected to establish the calibration function.

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Appendix A. Main characteristic of used data sets

Table A.1. Main characteristic of used data set for TSS (SD: Standard Deviation)

Name (number of samples)	Weather	Location	Min-mean-max Median – SD (mg/L) all samples	Min-mean-max Median – SD (mg/L) calibration samples	Min-mean-max Median – SD (mg/L) verification samples
INSA-AW (135)	All	WWTP	26–157–322 160–78	26–162–322 166–78	31–143–311 146–76
INSA-DW (78)	Dry	WWTP	26–166–389 188–84	26–166–389 177–80	54–215–362 222–82
INSA-WW (34)	Wet	WWTP	56–165–402 154–79	56–161–402 152.5–85	84–173–285 164–66
KWB-RW (51)	Wet	River	2–28–110 18–26	2–22–110 9–26	18–44–80 36–20
KWB-WW (59)	Wet	Sewer	4–163–590 110–155	4–114–430 58–123	11–276–590 270–165
PUJ-CCS (27)	Dry	Sewer	3–81–348 83–72	14–91–348 95–80	3–58–117 55–44
PUJ-WW (41)	Wet	Sewer	82–320–760 296–152	82–349–760 307–161	120–249–470 246–103
PUJ-RW (21)	Wet	River	2–38–133 19–43	2–38–133 20–44	3–37–105 15–45
TUG-Graz-AW (168)	All	Sewer	7–204–733 186–131	9–223–700 238–129	7–160–733 145–126
TUG-Graz-DW (85)	Dry	Sewer	9–211–656 232–127	9–207–656 229–129	10–221–422 246–124
TUG-Graz-WW (83)	Wet	Sewer	7–197–733 164–134	73–233–733 190–137	7–114–320 95–84
TUG-Linz-pcIN (24)	Dry	WWTP	157–317–562 330–95	203–351–562 342–79	157–232–380 204–81
TUG-Linz-pcOUT (24)	Dry	WWTP	71–98–124 101–16	71–102–124 105–15	73–86–101 87–10
TUG-Linz-pcALL (48)	Dry	WWTP	71–207–562 140.5–130	71–253–562 277–128	73–96–115 94.5–14

Table A.2. Main characteristic of used data set for COD total (SD: Standard Deviation).

Name (number of samples)	Weather	Location	Min-mean-max Median – SD (mg/L) all samples	Min-mean-max Median – SD (mg/L) calibration samples	Min-mean-max Median – SD (mg/L) verification samples
INSA-AW (136)	All	WWTP	59–333–717.5 290–180	74–379–717.5 387–184	59–227–570 207.5–114
INSA-DW (77)	Dry	WWTP	92–435–717.5 498–172	92–437–717.5 512–180	136–432–660 482–155
INSA-WW (36)	Wet	WWTP	74–251–581.5 248.5–123	92–256–570 248–118	74–241–581.5 249–140
KWB-RW (73)	Wet	River	23–38–107 30–18	23–32–67 29–10	27–51–107 39–26
KWB-WW (122)	Wet	Sewer	23–263–1085 140–270	23–159–1009 94–168	24–504–1085 464–307
PUJ-CCS (24)	Dry	Sewer	41–251–480 249–141	41–263–480 324–158	128–222–387 220–88
PUJ-WW (41)	Wet	Sewer	225–509–854 490–117	225–509–854 490–115	270–510–712 487–126
PUJ-RW (17)	Wet	River	9–52–167 38–43	9–57–167 35–48	10–40–79 40–27
TUG-Graz-AW (168)	All	Sewer	83–519–1 397 403–314	92–601–1 261 694–308	83–330–1 397 269–241
TUG-Graz-DW (85)	Dry	Sewer	131–697–1 261 774–263	164–707–1 261 774–260	131–674–1 160 768–274
TUG-Graz-WW (83)	Wet	Sewer	83–336–1 397 269–251	92–390–1 397 307–278	83–210–488 196–95
TUG-Linz-pcIN (24)	Dry	WWTP	380–692–906 733–148	668–771–906 765–71	380–500–707 492–101
TUG-Linz-pcOUT (24)	Dry	WWTP	267–432–507 462–78	340–463–507 488–52	267–359–464 307–86
TUG-Linz-pcALL (48)	Dry	WWTP	267–562–906 497–176	340–620–906 673–171	267–423–504 462–89

Table A.3. Main characteristic of used data set for COD dissolved (SD: Standard Deviation)

Name (number of samples)	Weather	Location	Min-mean-max Median – SD (mg/L) all samples	Min-mean-max Median – SD (mg/L) calibration samples	Min-mean-max Median – SD (mg/L) verification samples
INSA-AW (138)	All	WWTP	19–109–283 76–74	26–130–283 141–78	19–60–160 55–27
INSA-DW (79)	Dry	WWTP	31–153–283 163.5–72	31–168–283 186–76	39–119–190 141–49
INSA-WW (35)	Wet	WWTP	26–58–160 50–29	26–61–160 53–32	31–49–86 41–20
KWB-RW (56)	Wet	River	16–22–31 21.5–4	16–21–30 20–3	22–25–31 25–3
KWB-WW (121)	Wet	Sewer	12–59–233 44–44	12–49–233 39–40	20–82–176 73–45
PUJ-CCS (18)	Dry	Sewer	84–271–427 273–107	87–306–427 340–101	84–180–240 178–63
PUJ-WW (41)	Wet	Sewer	91–352–607 350–103	91–350–607 348–109	172–357–471 393–93
PUJ-RW (12)	Wet	River	4–36–63 37–17	4–42–63 41–18	17–25–33 25–9
TUG-Graz-AW (85)	All	Sewer	26–193–342 202–85	83–214–330 203–67	26–143–342 111–102
TUG-Graz-DW (74)	Dry	Sewer	79–212–342 203–72	83–203–330 202–65	79–233–342 243–84
TUG-Graz-WW (11)	Wet	Sewer	26–65–187 43–48	26–66–187 38–56	43–63–86 59–22
TUG-Linz-pcIN (24)	Dry	WWTP	120–182–274 176–36	156–194.5–274 182–84	120–153–181 156–19
TUG-Linz-pcOUT (24)	Dry	WWTP	140–181–231 180–21	160–185–231 183–18	140–173–204 175–26
TUG-Linz-pcALL (48)	Dry	WWTP	120–182–274 178–29	120–182–274 175–32	140–181–204 185–21

Appendix B. Values of applied criteria for each pollutant, method and data set.

Table B.1. Criteria for TSS calibrations. Cal. for calibration sub data sets and Ver. for verification sub data sets.

Method	Name of the data set	r^2		Nr^2_{01}		RMSE (mg/L)		$NRMSE_{01}$		NRMSE	
		Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.
LR	INSA-AW	0,8760	0,9211	1,00	1,00	27,42	28,18	1,00	1,00	0,0927	0,1004
	INSA-DW	0,9044	0,9308	1,00	0,00	24,51	44,10	1,00	0,09	0,0676	0,1428
	INSA-WW	0,9574	0,9479	0,00	0,92	17,12	14,49	0,43	1,00	0,0495	0,0722
	KWB-WW	0,8683	0,9149	0,00	1,00	43,88	81,54	0,24	0,51	0,1029	0,1408
	KWB-RW	0,5191	0,6724	0,00	1,00	17,84	12,38	0,14	1,00	0,1652	0,1997
	TUG-Graz-AW	0,4008	0,6214	0,00	0,01	98,91	93,43	0,15	0,51	0,1431	0,1287
	TUG-Graz-DW	0,6285	0,3620	0,00	0,00	77,08	107,15	0,01	0,39	0,1191	0,2601
	TUG-Graz-WW	0,4119	0,4444	0,00	0,73	103,99	122,77	0,06	0,70	0,1576	0,3931
	TUG-Linz-pcIN	0,8733	0,7519	0,00	1,00	42,62	33,20	0,05	0,48	0,0868	0,7905
	TUG-Linz-pcOUT	0,8635	0,9827	0,00	1,00	23,84	42,20	0,49	0,34	0,0664	0,1892
	TUG-Linz-pcALL	0,7876	0,6595	0,00	0,00	6,75	13,24	0,08	0,00	0,1274	0,4727
	PUJ-CCS	0,4615	0,6248	0,00	0,90	56,82	31,12	0,07	0,63	0,1706	0,2745
	PUJ-WW	0,5121	0,7449	0,08	1,00	109,81	38,46	0,17	1,00	0,2072	0,1261
	PUJ-RW-WW	0,2473	0,6215	0,00	0,43	36,82	41,85	0,02	0,53	0,2804	0,4089
EVO	INSA-AW	0,7632	0,8317	0,00	0,26	38,70	45,89	0,00	0,58	0,1308	0,1635
	INSA-DW	0,7224	0,9403	0,00	0,35	41,87	46,45	0,00	0,00	0,1154	0,1505
	INSA-WW	0,9576	0,9669	0,00	1,00	30,05	19,16	0,00	0,88	0,0869	0,0955
	KWB-WW	0,8683	0,9149	0,00	1,00	48,07	58,45	0,00	1,00	0,1127	0,1009
	KWB-RW	0,5512	0,6725	0,08	1,00	19,41	14,26	0,00	0,33	0,1797	0,2300
	TUG-Graz-AW	0,4008	0,6214	0,00	0,01	103,01	81,91	0,00	0,78	0,1491	0,1128
	TUG-Graz-DW	0,6285	0,3620	0,00	0,00	77,25	105,73	0,00	0,46	0,1194	0,2566
	TUG-Graz-WW	0,4119	0,4444	0,00	0,73	110,42	108,48	0,00	0,80	0,1673	0,3473
	TUG-Linz-pcIN	0,8733	0,7519	0,00	1,00	44,49	49,18	0,00	0,00	0,0906	1,1710
	TUG-Linz-pcOUT	0,8636	0,9826	0,00	0,99	35,20	24,97	0,00	1,00	0,0981	0,1120
	TUG-Linz-pcALL	0,7876	0,6599	0,00	0,00	6,91	9,70	0,00	0,49	0,1304	0,3464
	PUJ-CCS	0,4626	0,6253	0,00	0,90	57,82	36,85	0,00	0,33	0,1736	0,3251
	PUJ-WW	0,4942	0,7448	0,00	1,00	113,58	44,85	0,00	0,86	0,2143	0,1470
	PUJ-RW-WW	0,2466	0,6182	0,00	0,41	37,50	38,72	0,00	0,60	0,2856	0,3783
SVM	INSA-AW	0,8400	0,8000	0,68	0,00	31,00	70,00	0,68	0,00	0,1048	0,2494
	INSA-DW	0,8400	0,9500	0,65	0,72	33,00	43,00	0,51	0,13	0,0910	0,1393
	INSA-WW	0,9800	0,9600	0,53	0,97	13,00	16,00	0,57	0,96	0,0376	0,0797
	KWB-WW	0,9000	0,9000	0,47	0,87	36,00	83,00	0,70	0,48	0,0844	0,1434
	KWB-RW	0,9000	0,5900	1,00	0,00	8,00	13,00	1,00	0,78	0,0741	0,2097
	TUG-Graz-AW	0,6100	0,6200	0,84	0,00	80,00	115,00	0,85	0,00	0,1158	0,1584
	TUG-Graz-DW	0,6600	0,6200	0,30	0,65	77,00	95,00	0,02	1,00	0,1190	0,2306
	TUG-Graz-WW	0,7600	0,6100	0,59	1,00	67,00	79,00	0,42	1,00	0,1015	0,2529
	TUG-Linz-pcIN	0,9800	0,6900	0,86	0,71	18,00	16,00	0,70	1,00	0,0367	0,3810
	TUG-Linz-pcOUT	0,9800	0,9800	1,00	0,75	12,00	51,00	1,00	0,00	0,0334	0,2287
	TUG-Linz-pcALL	0,8800	0,7200	1,00	0,48	5,00	6,00	1,00	1,00	0,0943	0,2143
	PUJ-CCS	0,6100	0,6700	0,46	1,00	49,00	24,00	0,63	1,00	0,1471	0,2118
	PUJ-WW	0,7100	0,6700	1,00	0,59	92,00	62,00	1,00	0,47	0,1736	0,2033
	PUJ-RW-WW	0,9800	0,7900	1,00	1,00	5,00	21,00	1,00	1,00	0,0381	0,2052
PLS	INSA-AW	0,8295	0,8199	0,59	0,16	32,24	40,66	0,57	0,70	0,1090	0,1449
	INSA-DW	0,8140	0,9576	0,50	1,00	34,18	20,12	0,44	1,00	0,0942	0,0652
	INSA-WW	1,0000	0,7237	1,00	0,00	0,13	53,95	1,00	0,00	0,0004	0,2688
	KWB-WW	0,9353	0,8015	1,00	0,00	30,93	106,03	1,00	0,00	0,0725	0,1831
	KWB-RW	0,8301	0,6006	0,82	0,13	10,62	15,20	0,77	0,00	0,0983	0,2452
	TUG-Graz-AW	0,6491	0,7379	1,00	1,00	75,84	72,46	1,00	1,00	0,1098	0,0998
	TUG-Graz-DW	0,7351	0,7561	1,00	1,00	65,93	114,78	1,00	0,00	0,1019	0,2786
	TUG-Graz-WW	0,9972	0,0055	1,00	0,00	7,23	226,32	1,00	0,00	0,0110	0,7246
	TUG-Linz-pcIN	0,9971	0,5416	1,00	0,00	6,82	22,57	1,00	0,80	0,0139	0,5374
	TUG-Linz-pcOUT	0,9555	0,9720	0,79	0,00	16,28	26,87	0,82	0,93	0,0453	0,1205
	TUG-Linz-pcALL	0,8300	0,7857	0,46	1,00	6,11	8,59	0,42	0,64	0,1153	0,3068
	PUJ-CCS	0,7857	0,2355	1,00	0,00	43,77	43,20	1,00	0,00	0,1314	0,3812
	PUJ-WW	0,5811	0,5604	0,40	0,00	102,49	82,79	0,51	0,00	0,1934	0,2714
	PUJ-RW-WW	0,8514	0,4965	0,82	0,00	16,41	65,76	0,65	0,00	0,1250	0,6426

Table B.2. Criteria for COD total calibrations. Cal. for calibration sub data sets and Ver. for verification sub data sets.

Method	Name of the data set	r^2		Nr^2_{01}		RMSE (mg/L)		$NRMSE_{01}$		NRMSE	
		Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.
LR	INSA-AW	0,7938	0,8729	0,00	1,00	83,23	54,48	0,00	1,00	0,1293	0,1067
	INSA-DW	0,6601	0,4298	0,00	0,00	100,44	148,06	0,00	0,00	0,1605	0,2827
	INSA-WW	0,8482	0,9501	0,03	0,44	44,44	28,40	0,06	0,69	0,0929	0,0559
	KWB-WW	0,7905	0,8592	0,00	0,94	76,53	181,56	0,31	0,80	0,0776	0,1712
	KWB-RW	0,0016	0,5464	0,00	0,23	10,00	30,40	0,01	0,00	0,2253	0,3783

(continued on next page)

(*continued*)

Method	Name of the data set	r^2		Nr^2_{01}		RMSE (mg/L)		$NRMSE_{01}$		NRMSE	
		Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.
	TUG-Graz-AW	0,8109	0,8051	0,00	0,52	133,09	110,14	0,10	0,99	0,1138	0,0838
	TUG-Graz-DW	0,8128	0,8947	0,00	0,51	111,12	161,13	0,05	0,00	0,1013	0,1565
	TUG-Graz-WW	0,7712	0,1076	0,00	0,00	131,27	129,06	0,10	0,64	0,1006	0,3189
	TUG-Linz-pcIN	0,7731	0,7681	0,00	0,00	77,80	60,04	0,08	0,53	0,1217	0,2533
	TUG-Linz-pcOUT	0,3415	0,5355	0,00	0,00	53,02	187,36	0,56	0,00	0,2228	0,5730
	TUG-Linz-pcALL	0,8191	0,7863	0,00	1,00	21,28	53,53	0,04	1,00	0,1274	0,2722
	PUJ-CCS	0,9213	0,9403	0,01	1,00	41,72	40,41	0,20	0,89	0,0951	0,1561
	PUJ-WW	0,0018	0,5842	0,00	1,00	106,37	115,40	0,28	1,00	0,1774	0,2610
	PUJ-RW-WW	0,3328	0,0340	0,00	0,18	37,65	24,37	0,00	1,00	6,5697	11,7847
EVO	INSA-AW	0,9082	0,7853	0,90	0,00	56,80	84,19	0,82	0,06	0,0882	0,1649
	INSA-DW	0,8486	0,9638	0,86	1,00	63,31	29,73	0,99	1,00	0,1011	0,0568
	INSA-WW	0,8436	0,9438	0,00	0,32	47,22	43,43	0,00	0,12	0,0987	0,0856
	KWB-WW	0,7905	0,8592	0,00	0,94	89,33	233,54	0,00	0,00	0,0906	0,2202
	KWB-RW	0,0009	0,5419	0,00	0,21	10,10	27,74	0,00	0,32	0,2275	0,3452
	TUG-Graz-AW	0,8108	0,8050	0,00	0,52	137,87	110,02	0,00	1,00	0,1179	0,0837
	TUG-Graz-DW	0,8128	0,8947	0,00	0,51	112,83	156,72	0,00	0,10	0,1029	0,1522
	TUG-Graz-WW	0,7712	0,1076	0,00	0,00	142,83	120,64	0,00	0,73	0,1094	0,2981
	TUG-Linz-pcIN	0,7731	0,7681	0,00	0,00	80,84	84,63	0,00	0,00	0,1265	0,3571
	TUG-Linz-pcOUT	0,3415	0,5355	0,00	0,00	76,38	88,02	0,00	1,00	0,3210	0,2692
	TUG-Linz-pcALL	0,8191	0,7863	0,00	1,00	22,10	68,69	0,00	0,37	0,1323	0,3492
	PUJ-CCS	0,9205	0,9399	0,00	1,00	48,20	51,66	0,00	0,63	0,1098	0,1995
	PUJ-WW	0,0013	0,4444	0,00	0,75	132,68	115,51	0,00	1,00	0,2213	0,2612
	PUJ-RW-WW	0,3315	0,0343	0,00	0,19	37,67	24,63	0,00	0,82	6,5726	11,9103
SVM	INSA-AW	0,9200	0,8000	0,99	0,17	51,00	86,00	1,00	0,00	0,0792	0,1685
	INSA-DW	0,8800	0,8600	1,00	0,81	63,00	56,00	1,00	0,78	0,1006	0,1069
	INSA-WW	0,9700	0,9800	0,81	1,00	20,00	20,00	0,62	1,00	0,0418	0,0394
	KWB-WW	0,9000	0,8600	0,85	1,00	52,00	188,00	0,90	0,71	0,0528	0,1772
	KWB-RW	0,9900	0,4900	1,00	0,00	1,00	28,00	1,00	0,29	0,0225	0,3485
	TUG-Graz-AW	0,8800	0,7200	0,68	0,00	104,00	118,00	0,71	0,29	0,0890	0,0898
	TUG-Graz-DW	0,9000	0,8600	1,00	0,00	79,00	119,00	1,00	1,00	0,0720	0,1156
	TUG-Graz-WW	0,9600	0,2400	0,86	0,78	63,00	97,00	0,69	1,00	0,0483	0,2397
	TUG-Linz-pcIN	0,9200	0,8600	0,92	1,00	49,00	38,00	0,86	1,00	0,0766	0,1603
	TUG-Linz-pcOUT	0,7400	0,7400	1,00	1,00	35,00	167,00	1,00	0,20	0,1471	0,5107
	TUG-Linz-pcALL	1,0000	0,7600	1,00	0,79	4,00	73,00	0,92	0,19	0,0240	0,3712
	PUJ-CCS	1,0000	0,9400	1,00	1,00	15,00	36,00	1,00	1,00	0,0342	0,1391
	PUJ-WW	0,2700	0,5300	0,30	0,90	102,00	123,00	0,33	0,85	0,1702	0,2781
	PUJ-RW-WW	0,9200	0,1400	1,00	1,00	21,00	25,00	0,71	0,56	3,6643	12,0890
PLS	INSA-AW	0,9210	0,8068	1,00	0,25	51,55	76,52	0,98	0,30	0,0801	0,1499
	INSA-DW	0,8709	0,9634	0,96	1,00	64,26	43,79	0,97	0,88	0,1027	0,0836
	INSA-WW	0,9993	0,9269	1,00	0,00	3,07	46,74	1,00	0,00	0,0064	0,0921
	KWB-WW	0,9188	0,8476	1,00	0,00	47,80	168,95	1,00	1,00	0,0485	0,1593
	KWB-RW	0,6064	0,7385	0,61	1,00	6,29	22,17	0,42	1,00	0,1417	0,2759
	TUG-Graz-AW	0,9129	0,8827	1,00	1,00	90,38	121,23	1,00	0,00	0,0773	0,0923
	TUG-Graz-DW	0,8711	0,9285	0,67	1,00	92,58	145,07	0,60	0,38	0,0844	0,1409
	TUG-Graz-WW	0,9900	0,2773	1,00	1,00	27,57	185,37	1,00	0,00	0,0211	0,4580
	TUG-Linz-pcIN	0,9320	0,8503	1,00	0,89	44,02	57,82	1,00	0,57	0,0689	0,2440
	TUG-Linz-pcOUT	0,4091	0,6000	0,17	0,32	52,66	141,92	0,57	0,46	0,2213	0,4340
	TUG-Linz-pcALL	0,9979	0,6622	0,99	0,00	2,34	77,64	1,00	0,00	0,0140	0,3948
	PUJ-CCS	0,9598	0,8223	0,49	0,00	30,79	77,79	0,52	0,00	0,0702	0,3005
	PUJ-WW	0,8977	0,0315	1,00	0,00	38,49	167,00	1,00	0,00	0,0642	0,3776
	PUJ-RW-WW	0,9069	0,0100	0,98	0,00	14,24	25,79	1,00	0,00	2,4847	12,4710

Table B.3. Criteria for COD dissolved calibrations. Cal. for calibration sub data sets and Val. for verification sub data sets.

Method	Name of the data set	r^2		Nr^2_{01}		RMSE (mg/L)		$NRMSE_{01}$		NRMSE	
		Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.
LR	INSA-AW	0,8977	0,5981	0,26	0,46	24,66	57,72	0,33	0,00	0,0961	0,4095
	INSA-DW	0,8606	0,7502	0,19	0,49	28,20	38,27	0,20	0,66	0,1119	0,2535
	INSA-WW	0,7146	0,5460	1,00	1,00	16,64	12,72	1,00	1,00	0,1239	0,2292
	KWB-WW	0,7383	0,8211	0,27	1,00	20,40	17,91	0,32	1,00	0,0923	0,1148
	KWB-RW	0,2008	0,2285	0,19	0,44	2,82	3,17	0,11	1,00	0,1976	0,3339
	TUG-Graz-AW	0,3226	0,6233	0,00	1,00	54,85	74,22	0,15	0,70	0,2221	0,2349
	TUG-Graz-DW	0,3976	0,6401	0,00	0,86	49,75	75,61	0,12	0,78	0,2014	0,2875
	TUG-Graz-WW	0,7270	0,3815	0,00	0,53	27,17	24,40	0,00	0,86	0,1688	0,5675
	TUG-Linz-pcIN	0,1096	0,6865	0,29	0,62	28,85	16,87	0,81	0,87	0,1870	0,2623
	TUG-Linz-pcOUT	0,1158	0,4350	0,00	1,00	29,74	72,91	0,61	0,00	0,2520	1,1952
	TUG-Linz-pcALL	0,1020	0,9178	0,00	0,95	16,77	16,20	0,00	1,00	0,2363	0,2518
	PUJ-CCS	0,8063	0,7668	0,00	1,00	41,41	82,80	0,00	0,00	0,2958	0,5323
	PUJ-WW	0,0671	0,6856	0,08	0,84	100,89	58,45	0,05	1,00	0,1957	0,1956
	PUJ-RW-WW	0,8338	0,9069	0,00	1,00	6,24	14,82	1,00	0,35	0,1060	0,8896
EVO	INSA-AW	0,8700	0,4756	0,00	0,00	31,03	56,23	0,00	0,04	0,1209	0,3989

(*continued*)

Method	Name of the data set	r^2		Nr^2_{01}		RMSE (mg/L)		$NRMSE_{01}$		NRMSE	
		Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.	Cal.	Ver.
	INSA-DW	0,8392	0,7820	0,00	1,00	30,76	50,96	0,00	0,00	0,1220	0,3376
	INSA-WW	0,0705	0,0003	0,00	0,00	36,93	23,97	0,00	0,00	0,2750	0,4320
	KWB-WW	0,6699	0,7618	0,00	0,30	24,94	36,74	0,00	0,27	0,1128	0,2355
	KWB-RW	0,0141	0,0000	0,00	0,00	3,16	4,00	0,00	0,55	0,2218	0,4207
	TUG-Graz-AW	0,3226	0,6233	0,00	1,00	59,96	58,94	0,00	1,00	0,2427	0,1865
	TUG-Graz-DW	0,3976	0,6401	0,00	0,86	53,67	82,68	0,00	0,60	0,2173	0,3144
	TUG-Graz-WW	0,7270	0,3815	0,00	0,53	27,25	24,47	0,00	0,86	0,1693	0,5691
	TUG-Linz-pcIN	0,0360	0,4007	0,00	0,00	42,32	42,88	0,00	0,00	0,2743	0,6668
	TUG-Linz-pcOUT	0,2776	0,3762	0,21	0,80	48,48	16,75	0,00	1,00	0,4109	0,2745
	TUG-Linz-pcALL	0,1382	0,9500	0,04	1,00	16,74	19,10	0,00	0,85	0,2357	0,2969
	PUJ-CCS	0,8968	0,6375	0,68	0,37	36,20	62,47	0,32	0,50	0,2585	0,4016
	PUJ-WW	0,0241	0,8032	0,00	1,00	102,40	86,68	0,00	0,52	0,1986	0,2900
	PUJ-RW-WW	0,8372	0,9018	0,04	0,99	9,51	13,38	0,00	1,00	0,1615	0,8031
SVM	INSA-AW	0,9700	0,7400	0,94	1,00	13,00	21,00	0,94	0,98	0,0507	0,1490
	INSA-DW	0,9500	0,7200	1,00	0,00	18,00	44,00	1,00	0,36	0,0714	0,2915
	INSA-WW	0,1500	0,0900	0,12	0,16	31,00	20,00	0,29	0,35	0,2309	0,3605
	KWB-WW	0,8700	0,8200	0,78	0,99	16,00	20,00	0,63	0,92	0,0724	0,1282
	KWB-RW	0,9800	0,5200	1,00	1,00	0,00	5,00	1,00	0,00	0,0000	0,5263
	TUG-Graz-AW	0,6100	0,4600	0,54	0,39	42,00	84,00	0,52	0,50	0,1700	0,2658
	TUG-Graz-DW	0,7600	0,7100	0,72	1,00	33,00	67,00	0,62	1,00	0,1336	0,2548
	TUG-Graz-WW	0,9400	0,4500	0,78	1,00	23,00	18,00	0,17	1,00	0,1429	0,4186
	TUG-Linz-pcIN	0,2900	0,6600	1,00	0,56	28,00	18,00	0,86	0,83	0,1815	0,2799
	TUG-Linz-pcOUT	0,9000	0,1400	1,00	0,00	18,00	39,00	1,00	0,60	0,1525	0,6393
	TUG-Linz-pcALL	0,5600	0,7900	0,53	0,73	13,00	23,00	0,28	0,64	0,1831	0,3575
	PUJ-CCS	0,9400	0,5600	1,00	0,00	25,00	42,00	1,00	1,00	0,1786	0,2700
	PUJ-WW	0,1100	0,0700	0,15	0,00	102,00	90,00	0,01	0,46	0,1978	0,3011
	PUJ-RW-WW	0,9000	0,7700	0,85	0,78	9,00	14,00	0,16	0,72	0,1528	0,8403
PLS	INSA-AW	0,9767	0,7149	1,00	0,90	11,80	20,34	1,00	1,00	0,0460	0,1443
	INSA-DW	0,9320	0,7745	0,84	0,88	19,73	31,69	0,86	1,00	0,0783	0,2099
	INSA-WW	0,0710	0,0003	0,00	0,00	30,09	22,98	0,34	0,09	0,2241	0,4142
	KWB-WW	0,9261	0,7361	1,00	0,00	10,84	43,66	1,00	0,00	0,0491	0,2799
	KWB-RW	0,2519	0,0298	0,25	0,06	2,72	4,15	0,14	0,46	0,1909	0,4368
	TUG-Graz-AW	0,8575	0,3561	1,00	0,00	25,23	109,25	1,00	0,00	0,1021	0,3457
	TUG-Graz-DW	0,8990	0,2072	1,00	0,00	20,43	105,97	1,00	0,00	0,0827	0,4029
	TUG-Graz-WW	0,9990	0,3040	1,00	0,00	1,88	65,19	1,00	0,00	0,0117	1,5160
	TUG-Linz-pcIN	0,2774	0,8618	0,95	1,00	25,72	13,02	1,00	1,00	0,1667	0,2025
	TUG-Linz-pcOUT	0,2937	0,4162	0,23	0,94	28,99	71,84	0,64	0,02	0,2457	1,1777
	TUG-Linz-pcALL	0,9613	0,3581	1,00	0,00	3,49	35,13	1,00	0,00	0,0492	0,5461
	PUJ-CCS	0,9288	0,6021	0,92	0,20	27,39	59,03	0,85	0,58	0,1956	0,3795
	PUJ-WW	0,5799	0,2389	1,00	0,23	69,32	116,92	1,00	0,00	0,1345	0,3912
	PUJ-RW-WW	0,9119	0,2811	1,00	0,00	6,88	15,59	0,80	0,00	0,1168	0,9358

Appendix C. Supplementary material

Supplementary material related to this article can be found at http://dx.doi.org/10.1016/j.watres.2016.05.070.

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